

Madhur Gupta

Full Stack Developer & Spatial Data Engineer

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PROFESSIONAL EXPERIENCE

Qen Labs

Feb 2024 – Present

Full Stack Developer

Remote

- Engineered a high-throughput geospatial visualization engine using **Kepler.gl** and **deck.gl** to process and render datasets exceeding **1M+ spatiotemporal points**, cutting client memory overhead by **40%** via viewport-driven rendering.
- Developed automated ingestion pipelines and dynamic map interfaces for vector tiles (**GeoJSON**, **Uber H3**) across 10+ complex layers using **PostGIS** and spatial indexing.
- Optimized developer workflows by integrating local and cloud AI models (**Claude Code**, **Cursor**) for component scaffolding, unit test coverage, and code refactoring, accelerating sprint velocity by **30%**.
- Refactored monolithic frontend architectures into modular, reusable **TypeScript** hooks and managed global application state using **Redux Toolkit**.
- Configured cloud data storage pipelines on **Google Cloud Storage (GCS)** and deployed containerized backend microservices to **GCP Cloud Run** secured with **OAuth 2.0**.
- Streamlined local and production environments using **Docker**, cutting build and setup times by **50%**.

Professos

Jun 2023 – Aug 2023

Full Stack Intern

Remote

- Built core candidate matching and job application management modules for a **MERN stack** recruitment platform.
- Re-architected frontend UI layouts using **Tailwind CSS** and **React**, improving component modularity and web page loading speed across desktop and mobile devices.

KEY PROJECTS

Algorithm Visualizer & Multiplayer Platform

Live Demo

React, Node.js, Socket.io, Tailwind CSS

- Built a real-time WebSocket sync layer with **Socket.io** achieving sub-100ms latency for **50+ concurrent users** in multiplayer algorithmic puzzle sessions.
- Designed interactive step-by-step visualizers for Graph (Dijkstra, A*) and Sorting algorithms, executing smooth **60fps animations** using custom React hooks.
- Implemented a deterministic seed generator producing reproducible Sudoku boards and shareable match room IDs.

DevCleaner CLI Utility

GitHub Repository

Python, Concurrent Futures, CLI

- Developed an interactive command-line tool in Python to scan and purge heavy development dependency directories (`node_modules`, `.venv`, build caches) concurrently.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript (ES6+), Python, C++, SQL (PostgreSQL), HTML5/CSS3

Frontend & Geospatial: React, Next.js, Redux, Deck.gl, Kepler.gl, Uber H3, PostGIS, Tailwind CSS

Backend & Cloud: Node.js, Express, GCP (Cloud Run, GCS), Docker, MongoDB, Socket.io, REST APIs

Developer Tools: Git, Linux/Bash, VS Code, Postman, Claude Code, Cursor, TrueNAS/Containers

EDUCATION

Indian Institute of Information Technology (IIIT), Kota

Dec 2020 – June 2024

Bachelor of Technology in Electronics and Communication Engineering

Kota, Rajasthan

- Relevant Coursework:** Data Structures & Algorithms, Object-Oriented Programming, Operating Systems, Database Management Systems (DBMS), Computer Networks, Software Engineering.

ACHIEVEMENTS & RESEARCH

- ACM Publication:** Lead Author of "MapYog - Intelligent Spatiotemporal Data Explorer" (ACM, 2024), presenting research on high-scale spatial data visualization architectures.
- Codeshastra 8.0 Winner:** Secured 1st place by building a geolocation-verified facial attendance prototype with real-time API verification.
- GATE CS 2024:** Qualified the GATE Computer Science & Information Technology examination.
- Competitive Programming:** Solved **1000+ algorithmic problems** across LeetCode and Codeforces covering Data Structures, Dynamic Programming, and Graph Algorithms.